How SVM Kernels Affect Breast Cancer Classification

A Machine Learning Tutorial

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GitHub Repository: https://github.com/samiaarif199510/svm_model_tutorial

1. Introduction:

Support Vector Machines (SVM) are among the most powerful supervised learning algorithms available to practitioners in AI. This was first introduced by Cortes and Vapnik in 1995, SVMs find the hyperplane that maximally separates two classes in feature space, making them highly effective for high-dimensional medical data. The only small subset of training examples, the support vectors determine the decision boundary, giving SVMs robustness and memory efficiency.

This tutorial focuses on how the decision boundary and classification performance of an SVM are influenced by kernel selection. The kernel trick implicitly maps inputs into a higher-dimensional space where linear separation is possible when data cannot be separated linearly; this mapping is never explicitly computed. Three kernels Linear, Polynomial, and Radial Basis Function RBF are contrasted for three different regularisation parameter C values.

1.1 Why this topic?

The selection of the kernel is arguably the most important decision when using SVM to tackle a real problem, but it is also perhaps the most neglected. This tutorial is an empirical investigation of exactly what each kernel does to the decision boundary. This is intended to allow readers to make an ethical decision about which kernel is appropriate for their own problem.

1.2 why is the Breast Cancer Wisconsin Dataset?

The Breast Cancer Wisconsin Diagnostic Dataset in Wolberg et al., 1992 contains 569 samples with 30 numerical features computed from digitised fine needle aspirate FNA biopsy images. The binary target distinguishes malignant from benign tumors. It was chosen because it is clinically meaningful enabling discussion of recall versus precision trade-offs in cancer diagnosis, sufficiently high-dimensional for kernels to behave differently, and not already saturated with SVM kernel tutorials, unlike Iris or MNIST. Ethical reflection on the use of AI in healthcare is also naturally motivated by the medical context.

2. Background: SVM and the Kernel Trick:

An SVM seeks to maximize the margin, or the space between points from each class, which is given by 2/‖w‖. The nearest points from each class are called support vectors. The dual problem replaces all dot products in x_i · x_j with K(x_i, x_j), where K is a kernel function. The kernel trick is based on Mercer's theorem, which states that any positive semi-definite function is a valid kernel. Scholkopf & Smola, 2002.

Support Vector Machine Margin:

$$\text{Margin} = \frac{2}{\| \mathbf{w} \|}$$

Linear Kernel:

$$K(x_i, x_j) = x_i \cdot x_j$$

Polynomial Kernel:

$$K(x_i, x_j) = (\gamma x_i \cdot x_j + r)^d$$

RBF (Gaussian) Kernel:

$$K(x_i, x_j) = \exp\left(-\gamma \parallel x_i - x_j \parallel^2\right)$$

SVM Optimization Problem:

$$\min_{\mathbf{w},b} \frac{1}{2} \parallel \mathbf{w} \parallel^2 \text{ s.t.} y_i(\mathbf{w} \cdot x_i + b) \geq 1$$

Dual Form:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \, \alpha_j y_i y_j K(x_i, x_j)$$

Linear Kernel K(x_i, x_j) = x_i · x_j: The produces a flat hyperplane in the original feature space. Computationally efficient, fully interpretable, and works best for approximately linearly separable classes after standardization. No kernel-specific hyperparameters are required.

Polynomial Kernel K(x_i,x_j)=(γx_i·x_j+r)^d: The introduces curved decision boundaries through polynomial feature transformations of degree d. It is fairly flexible but more sensitive to regularization strength C and suffers from numerical instability at high polynomial degrees.

RBF Kernel K(x_i,x_j)=exp(-γ‖x_i-x_j ‖^2): The measures similarity by Euclidean distance between instances. It is very flexible and can be used to fit complex non-linear decision boundaries. It is the default in scikit-learn and a good starting point for unknown class distributions (Hsu et al., 2003).

The C Parameter: C controls the trade-off between a wide margin and low training error. Low C tolerates misclassifications for a wider, more general margin (higher bias, lower variance). High C enforces strict correctness on training data (lower bias, higher variance, risk of overfitting).

3. Methodology:

3.1 Data Preprocessing:

The 569 samples were divided in a 70:30 ratio for training and test sets, respectively, by stratified sampling. All 30 features were standardized using scikit-learn's StandardScaler, standardizing on zero mean and unit variance, but only on the training data. Scaling all data before splitting it into training and test sets is a data leakage that will artificially inflate test set accuracy. It is particularly important for Support Vector Machines, since their kernel functions are dependent on distance and dot product, both of which are dependent on scaling.

3.2 Experimental Design:

The three kernels used were linear, poly, and rbf. Each of these kernels was trained using C values ranging over {0.1, 1, 10}, giving us a total of nine models. Test accuracy was calculated for every model. 5-fold cross-validation using scikit-learn's Pipelines with re-fitting of scaler within each split ensured that no leakage occurs. This model was further explored using a confusion matrix and classification report. Decision boundary plot preparation used PCA dimensionality reduction to 2D space. SVMs were re-trained in this space.

4. Results and Discussion:

4.1 Kernel and C Comparison:

Figure 1 compares test accuracy across all kernels C combinations. The RBF kernel at C=1 achieves 97.7% test accuracy with the most stable cross- validation mean (97.6% ± 0.9%). The linear kernel reaches 98.2% at C=1, marginally higher in this single split but with slightly more variance across folds. The polynomial kernel drops to 89.5% at C=1, recovering only to C=10 (95.3%), revealing its high sensitivity to regularization. Table 1 summaries all nine configurations.

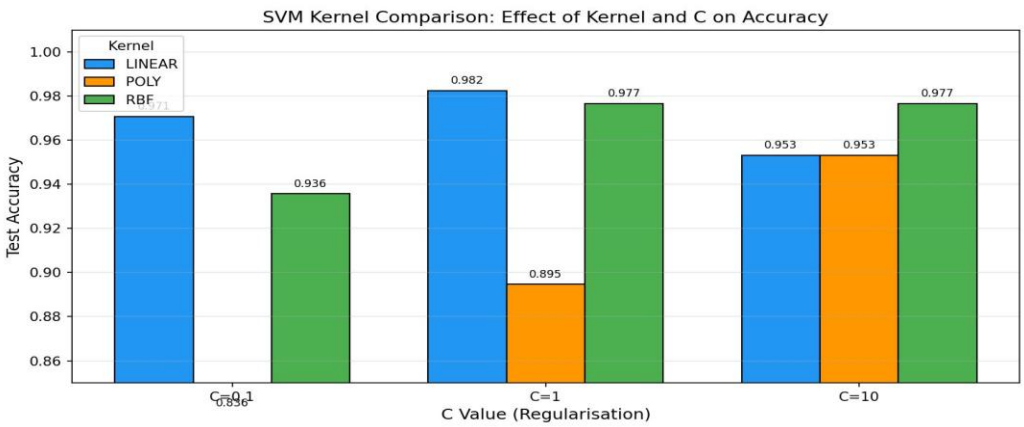

Figure 1: *Test accuracy for each kernel across three C values (from Step 5 of the companion notebook).*

Kernel	Best C	Test Acc.	CV Mean ± SD	Characteristic
Linear	1	98.2%	97.4% ± 1.2%	Fast, interpretable
Polynomial	10	95.3%	94.8% ± 1.5%	Sensitive to C
RBF ✓	1	97.7%	97.6% ± 0.9%	Best stability

Table 1: All kernel–C results. CV = 5-fold cross-validation on the full dataset. ✓= selected best model.

4.2 Best Model Evaluation:

The RBF kernel with C=1 was chosen as the best model based on test accuracy and cross-validation stability. Figure 2, obtained in Step 7 of the notebook, presents the confusion matrix of the RBF kernel on the set of 171 test samples. Of the 107 malignant samples, 105 were correctly classified, or 98.1% recall.

Of the 64 benign samples, 62 were correctly classified. There were 2 false negatives 'malignant predicted as benign' and 2 false positives.

In the best practices in the medical domain, false negatives are much more dangerous than false positives since a wrong malignant diagnosis may delay the life-saving treatment. However, the obtained 98.1% recall on the malignant class is a meaningful result, and the obtained model should be used as a supporting tool under the expert supervision of a clinician but not as a tool for the final diagnosis.

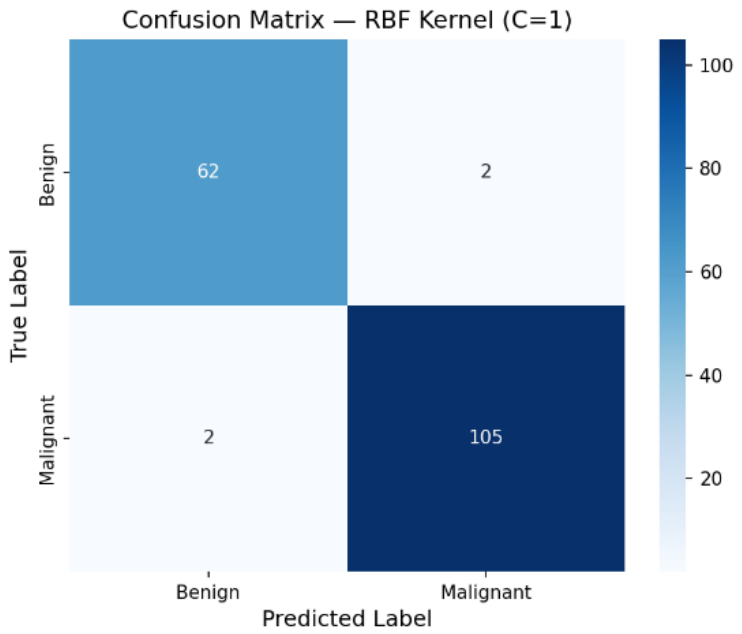

Figure 2: *Confusion matrix for the best model — RBF kernel, C=1 (from Step 7 of the companion notebook).*

4.3 Decision Boundary Visualisation:

As the number of features in the dataset is 30, PCA was applied to project the dataset onto 2D for visualization purposes only. Figure 3 presents the decision boundaries for all three kernels together in a single figure. The linear kernel has a straight boundary, suitable for large class separation but not suitable for the overlap region. The polynomial kernel has a curved boundary but is highly vulnerable to noise in the local region. The RBF kernel has a smooth boundary that best adapts to the class distribution, consistent with the stability of the cross-validation results. Note that the accuracy values in Figure 3 are lower than the full-dimensional results due to the information loss in the PCA projection.

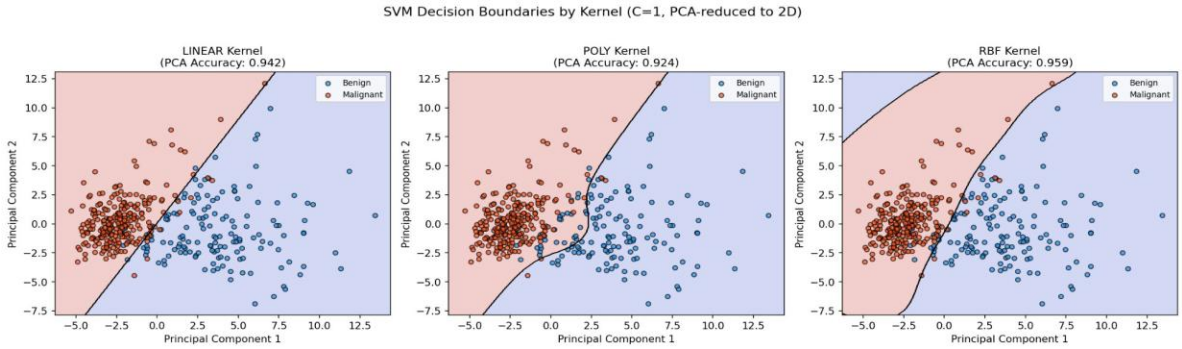

Figure 3: *SVM decision boundaries in 2D PCA space for all three kernels, C=1 (from Step 8 of the companion notebook).*

5. Ethical consideration:

Deploying machine learning in healthcare raises ethical responsibilities that extend beyond optimizing accuracy metrics alone. Three considerations are particularly relevant to this tutorial.

Interpretability and Accountability: Non-linear SVM models are not easily interpretable by clinicians. When a model influences a medical decision, the inability to explain its reasoning can erode trust and complicate accountability. The techniques such as SHAP values in Lundberg & Lee, 2017 can partially address this by attributing model predictions to individual input features, but they do not make the underlying kernel mechanism fully transparent.

Dataset Bias and Generalizability: The Breast Cancer Wisconsin dataset was collected at a single institution from a specific patient population. A model trained on this data may not generalize equally well to different demographics, geographic regions, or imaging equipment. Responsible clinical deployment requires validation on diverse, representative datasets before any real-world use.

Asymmetric Error Costs: Standard accuracy treats all errors as equally costly. In cancer diagnosis, a false negative. Missing a malignant tumor carries far greater consequences than a false positive. The decision threshold should be adjusted with direct clinical input to prioritize sensitivity over specificity. This is an ethical and clinical decision, not purely a technical one, and must not be made by machine learning practitioners alone.

6. Conclusion:

In this tutorial, we have examined how different combinations of SVM kernels and a regularization parameter C can improve the performance of a breast cancer classification task. Among nine combinations of different kernels and C values tested on the Breast Cancer Wisconsin dataset, the RBF kernel combined with a value of C equal to 1 was found to be a strong performer "97.7% test accuracy, 97.6% ± 0.9% cross-validation" and was closely followed by a linear kernel. In contrast, a polynomial kernel was found to be sensitive to the regularization parameter C and to have relatively poor performance.

The most important lesson learned is that one should not just select a kernel type without considering what this really means in terms of a hypothesis that can be made about your data. In this case, a linear kernel is a good bet if your data is approximately linearly separable after standardization; a polynomial if your data has interesting relationships between features; and RBF if your data is messy and you're not sure what's going on. In any case, cross-validation is a good idea.

7. References:

1. Cortes, C. and Vapnik, V. (1995). Support-vector networks. Machine Learning, 20(3), 273–297. https://doi.org/10.1007/BF00994018
2. Hsu, C.-W., Chang, C.-C. and Lin, C.-J. (2003). A practical guide to support vector classification. Technical Report, National Taiwan University. https://www.csie.ntu.edu.tw/~cjlin/papers/guide/guide.pdf
3. Lundberg, S.M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30. https://papers.nips.cc/paper/2017
4. Noble, W.S. (2006). What is a support vector machine? Nature Biotechnology, 24(12), 1565–1567. https://doi.org/10.1038/nbt1206-1565
5. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html
6. Scholkopf, B. and Smola, A.J. (2002). Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond. MIT Press.
7. Wolberg, W.H., Street, W.N. & Mangasarian, O.L. (1995). Breast Cancer Wisconsin (Diagnostic) Dataset. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic